Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

July 27, 2026

Dear Respondent,

Greetings!

We are fourth-year Bachelor of Science in Aviation Information Technology (BSAIT) Section 2 students from the National Aviation Academy of the Philippines, conducting our capstone project entitled: **"A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay"**

As part of the system evaluation, we respectfully invite you to evaluate the developed system. Your honest responses will help determine the quality and acceptability of the system and will serve as valuable input for improving its functionality and overall performance. Your participation is voluntary. You may decline to answer any question or withdraw from the evaluation at any time without any penalty.

Respectfully,

Charmaine Rose G. Angeles
Researcher

Kylane C. Gravino
Researcher

Melody D. Sedanto
Researcher

Noted by:

RICHARD NIEL B. PERALTA, MSIT
Capstone Adviser

ASSOC. PROF. MARY ANN ABALLIAR-VISTA, MEAM
Academic Adviser

In accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), all information collected through this questionnaire will be treated with the utmost confidentiality and will be used solely for academic and research purposes. Your identity will remain anonymous, and the information you provide will not be disclosed to any unauthorized person or organization. The collected data will be securely stored and disposed of after the study's completion.

By completing this questionnaire, you indicate your voluntary consent to participate in this study.

Consent Declaration

☑ I have read and understood the Data Privacy and Confidentiality Statement.

☑ I voluntarily agree to participate in this research study.

Thank you for your valuable time and participation.

Capstone Title: "A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay"

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Develop and implement a centralized digital job order entry and management system that captures customer information, service specifications, and transaction records, thereby reducing average service record retrieval time from 15-30 minutes to less than 5 minutes.
2. Implement and evaluate a machine learning model for predicting service completion dates with a target accuracy of 85% or higher compared to manual estimation methods.
3. Generate a real-time analytics dashboard that generates automated daily, weekly, monthly, quarterly, and annual reports, reducing manual report preparation time from 4-6 hours monthly to less than 1 hour.
4. Evaluate the effectiveness of the developed system based on the ISO 25010 software quality standards, specifically in terms of:
    - 4.1 Functional suitability;
    - 4.2 Performance efficiency;
    - 4.3 Compatibility;
    - 4.4 Interaction Capability;
    - 4.5 Reliability;
    - 4.6 Security;
    - 4.7 Maintainability;

4.8 Flexibility; and

4.9 Safety;

Additionally, the study will assess data accuracy and prediction reliability of the machine learning component.

5. Measure the level of user acceptance of the developed system among the staff and management of Shoelotskey using the Technology Acceptance Model (TAM), through a survey questionnaire measuring perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.
6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.

Survey Questionnaire

Part I. Respondent Profile

Instructions: Please provide the following information by filling in the data or putting a (✓) next to your response.

Name: (Optional) Jheo Enndi Corona

Role: ☑ IT Expert ☐ IT Practitioner

Position/Job Title: Software Developer/ Automation Specialist

Years of IT Experience:

☑ 5–7 years ☐ 8–10 years ☐ More than 10 years

Part II. System Evaluation (ISO 25010)

Instructions: To indicate your agreement level that best describes how you evaluate the system based on ISO 25010 software quality characteristics, put a (✓) in the box that is next to each statement below using the coded scale for your response.

Scale	Interpretation	Description
4	**Highly Acceptable**	The system fully meets the evaluation criterion and performs exceptionally well.
3	**Acceptable**	The system meets the evaluation criterion and performs satisfactorily.
2	**Less Acceptable**	The system partially meets the evaluation criterion and requires improvement.
1	**Not Acceptable**	The system does not meet the evaluation criterion and requires significant improvement.

CRITERIA	4	3	2	1
Functional Suitability- *This section evaluates whether the system provides the necessary functions to perform daily tasks.*				
1. The system provides the necessary functions for managing job orders, customer information, inventory, reports, and release schedules.	✓			
2. Each system module produces the expected result based on the user's operation.		✓		
3. The Machine Learning component uses a Random Forest Regressor to analyze job order data, identify service patterns, and generate a predicted release date for operational decision-making.		✓		
4. The system's data analytics functions provide information for monitoring sales, services, and business performance.	✓			
5. The system supports the complete job order workflow, including New Order, On-Going, For Release, and Claimed statuses.	✓			
Performance Efficiency- *This section evaluates whether the system performs tasks quickly and efficiently.*				
1. The system retrieves database records and loads pages within an acceptable response time.		✓		
2. Job order processing and saving transactions are completed without noticeable delays.		✓		
3. Sales and expense reports can be generated for daily, weekly, monthly, quarterly, and annual periods within an acceptable response time.	✓			
4. The Machine Learning component generates a predicted release date from valid input data without noticeably delaying the transaction workflow.		✓		
5. Navigation between system modules occurs without noticeable delay, supporting efficient single-page application (SPA) routing.	✓			
Compatibility- *This section evaluates whether the system works properly with other system components and supported platforms.*				
1. The system operates properly and maintains its functional layout across commonly used web browsers.	✓			
2. Modifications made to existing records, such as job order status changes, are immediately reflected in all related system modules.	✓			
3. Information entered in one module is correctly synchronized across other system modules.	✓			
4. The Machine Learning output is correctly integrated into the Job Order module to display the predicted release date.		✓		
5. The system's data analytics functions retrieve and consolidate transaction data to display current operational metrics.		✓		

CRITERIA	4	3	2	1
Interaction Capability- *This section evaluates whether the system is easy to learn, understand, and use.*				
1. The system's menus, icons, and buttons are clearly labeled and easy to recognize.	✓			
2. The system's job order, inventory, service, and user management features are easy to learn.	✓			
3. The system provides validation messages and confirmation prompts to help prevent input errors and unintended actions.	✓			
4. The layout and navigation make it easy to move between different modules.	✓			
5. Frequently used tasks can be completed with minimal steps.	✓			
Reliability- *This section evaluates whether the system performs consistently and operates without failures.*				
1. The system performs consistently without unexpected interruptions during operation.	✓			
2. The system handles invalid or incomplete input without unexpected failure.		✓		
3. Transactions are processed without data loss or unintended changes to existing records.	✓			
4. The system supports manual PostgreSQL database backups through the Heroku platform to facilitate data recovery.	✓			
5. The Machine Learning component consistently generates a predicted release date when provided with a complete and valid job order input.	✓			
Security- *This section evaluates whether the system protects data from unauthorized access.*				
1. The system uses JSON Web Token (JWT) authentication with bcrypt password hashing, enforces a three-attempt login limit with a 15-minute lockout, and automatically terminates inactive sessions after 30 minutes, addressing OWASP A07: Identification and Authentication Failures.	✓			
2. The system enforces Role-Based Access Control (RBAC) for Staff and Owner roles and prevents direct access to unauthorized modules through URL requests, addressing OWASP A01: Broken Access Control.	✓			
3. The system uses parameterized queries and input validation to prevent SQL injection and invalid data submission, addressing OWASP A03: Injection.	✓			
4. The system records critical user activities and system events in audit logs, including actions, user information, timestamps, and relevant changes, addressing OWASP A09: Security Logging and Monitoring Failures.	✓			
5. The system uses HTTPS with SSL/TLS encryption to protect data transmitted between the client and application server, addressing OWASP A02: Cryptographic Failures.	✓			

CRITERIA	4	3	2	1
Maintainability- *This section evaluates whether the system is easy to maintain, update, and improve.*				
1. The system provides clear error messages identifying the specific field and cause of invalid input to assist troubleshooting.	✓			
2. The system allows incorrect data entries to be corrected without affecting completed transaction records.	✓			
3. The system remains functional when existing records, services, or configurations are modified.		✓		
4. The system logs administrative actions and critical errors to facilitate system monitoring and maintenance.	✓			
5. The system's modules can be updated or modified without disrupting the job order workflow.		✓		
Flexibility- *This section evaluates whether the system can adapt to future changes and business needs.*				
1. The system supports different user roles, such as Staff and Owner, with appropriate access permissions and functions.	✓			
2. The system supports different types of shoe services within the same workflow.	✓			
3. The system handles different customer transactions, such as single-service and multi-service orders within the same workflow.	✓			
4. The system accommodates different payment methods and payment statuses within the same transaction workflow.	✓			
5. The system allows new services to be added without affecting existing transactions.	✓			
Safety*- This section evaluates whether the system helps prevent errors and protects important data.*				
1. The system helps prevent accidental loss or deletion of important records.	✓			
2. The system requires explicit user confirmation before executing critical data deletions or modifications.		✓		
3. The system enforces mandatory field requirements before allowing a transaction to proceed.		✓		
4. The system aborts transactions that violate defined business logic or validation rules.	✓			
5. The system prevents duplicate submissions during transaction processing.	✓			

PART III. Comments and Suggestions

Instructions: Please provide your comments and recommendations regarding the developed system.

1. **Which feature or component of the system do you think performs best? Why?**

The job order form performs best. The customer name field automatically suggests existing customer records as you type, so returning customers can be selected in one click instead of re-entering their details. This speeds up encoding and reduces typing errors.

2. **What weaknesses or issues did you observe during your evaluation?**

No major functional issues encountered. Minor observations: the predicted release date did not generate ("ML: unavailable"), and currency values display inconsistently (₱912.5 instead of ₱912.50).

3. **What improvements do you recommend to enhance the system's quality in terms of functionality, performance, security, maintainability, or other software quality characteristics?**

Standardize currency display to two decimals and fix the negative pending payments value. Ensure the predicted release date generates or shows a clear fallback message.

4. **Are there additional features or enhancements that you would recommend for future versions of the system?**

Add SMS or email notifications when an order is ready for pickup, and low-stock alerts in Inventory Management. A customer history view showing past orders per client would also help.

5. **Other comments or suggestions:**

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"We assure you that your responses will be treated with the utmost confidentiality and used strictly for academic purposes."

Signature of the Respondent

Date: 10/09/2026